Since A is an $n \times n$ matrix, the columns of A form a basis of $\mathbb{R}^n$ is equivalent to to dim Col A = n. (from defn.) $\therefore$ (a) $\Leftrightarrow$ (m) $\Leftrightarrow$ O.

By defn. of rank, dim Col A = n is equivalent to rank A = n.

$\therefore$ (a) $\Leftrightarrow$ (o) $\Leftrightarrow$ (p).

By Theorem 14, rank A + dim Nul A = n. If rank A = n, then dim Nul A = 0.

& The vice-versa is also true. $\therefore$ (a) $\Leftrightarrow$ (p) $\Leftrightarrow$ (r)

By Defn. of dimension, dim Nul A = 0 $\Leftrightarrow$ Nul A = $\{0\}$.

$\therefore$ (a) $\Leftrightarrow$ (q) $\Leftrightarrow$ (r). □

PRACTICE PROBLEMS

1. Determine the dimension of the subspace H of $\mathbb{R}^3$ spanned by the vectors v_1, v_2 and v_3. (First, find a basis for H.)

$$v_1 = \begin{bmatrix} 2 \\ -8 \\ 6 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ -7 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ 6 \\ -7 \end{bmatrix}$$

Solution: By defn, a basis for H is a set of linearly independent vectors which spans H.

Let $A = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} 2 & 3 & -1 \\ -8 & -7 & 6 \\ 6 & -1 & -7 \end{bmatrix}$. By defn, Span$\{v_1, v_2, v_3\}$ = Col A.

By Theorem 13, the no. of pivot columns of A form a basis for the column space of A.

$$\begin{bmatrix} 2 & 3 & -1 \\ -8 & -7 & 6 \\ 6 & -1 & -7 \end{bmatrix} \xrightarrow[+4 \times \text{Row1}]{\text{Row2} = \text{Row2}} \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 6 & -1 & -7 \end{bmatrix} \xrightarrow[-3 \times \text{Row1}]{\text{Row3} = \text{Row3}} \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & -10 & -4 \end{bmatrix}$$

$$\xrightarrow[+2 \times \text{Row2}]{\text{Row3} = \text{Row3}} \begin{bmatrix} \boxed{2} & 3 & -1 \\ 0 & \boxed{5} & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

pivot columns. $\therefore$ Basis for Col A = $\{v_1, v_2\}$. By Defn, Dim H = 2.

It's easy to verify that the final matrix is in echelon form and Columns 1 and 2 are

2. Consider the basis $B = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ for $\mathbb{R}^2$. If $[x]_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, what is x?